Spatial–Spectral Fusion BiFormer: A Novel Dynamic Routing Approach for Hyperspectral Image Classification

Yiqun Wang, Lina Yang, *Member, IEEE*, Thomas Xinzhang Wu, *Senior Member, IEEE*, Kaiwen Tang, and Wanxing Zha

***Abstract*— Hyperspectral image (HSI) classification categorizes each pixel into specific land-cover types. While convolutional neural networks (CNNs) and Transformers have demonstrated commendable performance in HSI classification, they still face limitations in extracting high-resolution spatial and dense semantic spectral information from HSI data. This paper proposes a novel method with a dynamic routing mechanism called spatial–spectral fusion BiFormer (SSFBF), which leverages the Transformer's capability to model long-range dependencies with dynamic routing attention while incorporating CNN's local receptive fields for fine-grained semantic features in HSIs. Based on an "encoder–decoder" architecture, SSFBF employs bi-level routing attention (BRA) to capture high-resolution feature representations in the spatial domain and single-level spectral routing attention (SSRA) to extract discriminative semantic information in the spectral domain. Additionally, we develop a spatial–spectral position feedforward network (SSposFFN) with CNNs to mitigate local feature loss induced by attention, replacing the traditional MLP layer following BRA and SSRA. Furthermore, a novel spatial–spectral dynamic fusion (SSDF) module is designed to integrate features in both the spatial and spectral domains. Various experiments on four HSI datasets have demonstrated that our SSFBF approach exhibits superior versatility on HSI datasets and outperforms the state-of-the-art HSI classification methods.**

***Index Terms*— Attention, convolutional neural networks (CNNs), dynamic feature fusion, hyperspectral image (HSI) classification, Transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) have been widely used in remote sensing due to their rich spectral information and high resolution [1]. They include hundreds of consecutive spectral bands that can retain more physical and chemical properties of the various materials [2]. Researchers have explored a series of applications of HSIs, including image classification [3], target detection [4], denoising [5], and unmixing [6]. As a vital technology in HSI processing, HSI classification categorizes each pixel in HSIs, which is crucial in utilizing land-cover information.

Early research on HSI classification relied on classic machine learning methods [7], [8], [9]. However, these methods depend heavily on preexisting knowledge and handcrafted features. More recently, deep learning methods have been proposed in HSI classification. Many backbone networks have been proved to perform well in HSI classification [10], [11], [12], [13], [14], [15]. Among them, convolutional neural networks (CNN) have dominated the field due to their powerful convolution kernel's capabilities of local pattern extraction in limited HSI samples. Chen et al. [16] and Hamida et al. [17] proposed 2-D and 3-D CNN-based models to extract compelling spectral–spatial features. Roy et al. [18] proposed HybridSN, which utilized 3-D CNN for spatial–spectral features and subsequent 2-D CNN for more spatial representation. In addition, a series of convolution-based methods have been proposed using multiscale training, attention modules, and feature fusion. In [19], an enhanced multiscale feature fusion network (EMFFN) was designed to obtain a larger respective field for long-ranged information and multiscale features. In [20], cross-spatial attention is proposed to generate spectral–spatial features from HSIs. Zheng et al. [21] designed a rotation-invariant attention network (RIAN) to replace CNNs for rotation-invariant features from HSIs. Yang et al. [22] used the three-branch to obtain multiscale features and the concatenation module to fuse them. Li et al. [23] introduced a dual-single-channel 3-D CNN to extract spectral–spatial information. Despite the remarkable performance of CNN-based methods in HSI classification, several limitations remain as follows.

1) The CNN's receptive fields may be insufficient for global feature representation in HSIs.
2) Most CNN-based methods suffer from the computational complexity paradox: shallow layers with low complexity fail to extract sufficient features, while deeper layers lead to prohibitive computational costs.

In addition to the CNN, the Transformer exhibits significant potential [24], [25]. Due to its long-distance sequence inputs and multihead self-attention (MHSA), the Transformer can efficiently extract long-range dependencies from the global context. Hong et al. [26] initially developed Spectralformer.

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Mei et al. [27] proposed a Group-Aware Hierarchical Transformer (GAHT) confining MHSA to the local spatial–spectral context. Xu et al. [28] proposed a spatial–spectral 1-D Swin Transformer (SS1DSwin) to reveal local and hierarchical spatial–spectral relationships. Qiao et al. [29] proposed a multiscale neighborhood attention Transformer (MSNAT), which emphasized the neighborhood pixels and extracted multiscale spatial information using different local window sizes. Zhou et al. [30] developed an RNN-Transformer (RT) architecture to feature the multiple sequential characters in HSI patches. Furthermore, they designed a segmented recurrent Transformer to overcome sequential data challenges in RNNs and Transformers [31]. However, these methods still suffered some problems, as follows.

1) The Transformer overly relies on long-range dependencies and neglects the local features between neighboring HSI pixels, further limiting its performance.
2) The calculation of MHSA often results in colossal computing consumption, especially when dealing with high-dimensional feature spaces in HSIs.

In recent years, methods that combined CNNs and Transformers have achieved state-of-the-art results. Sun et al. [32] proposed a spectral–spatial feature tokenization Transformer (SSTFF) method adopting 2-D and 3-D CNNs for shallow features and a Transformer encoder for feature learning. Yang et al. [33] introduced a Transformer containing a spectral-adaptive 3-D convolution projection module and a convolution permutator. Qiu et al. [34] developed a Transformer backbone incorporating spatial attention and a spectral feature extraction module with CNNs. Ouyang et al. [35] developed a CNN-based token generator for multigranularity semantic tokens and a Transformer encoder for global dependencies. Roy et al. [36] introduced a morphological Transformer with spatial–spectral morphological convolution in conjunction with attention. Sun et al. [37] used CNNs to extract shallow features and transform them into memory tokens as the Transformer's input, enhancing the feature's abundance. However, for the above structures, the simple branch for shallow spectral features fails to adequately capture the global dependencies and local details in the spectral domain, impairing generalization to different HSI datasets. Moreover, the lack of cross-layer fusion mechanisms between the spatial and spectral branches leads to the loss of high-level semantic features in both spatial–spectral domains during the fusion process.

To overcome these issues, inspired by the above studies and [38], [39], we propose a novel CNN–Transformer-based network called spatial–spectral fusion BiFormer (SSFBF). The SSFBF can extract high-grained features in HSIs from spatial and spectral perspectives, respectively, and fuse them to capture informative spatial–spectral representations. First, inspired by the efficient bi-level routing attention (BRA), we develop dynamic routing attention called single-level spectral routing attention (SSRA) in the spectral domain and build an "encoder–decoder" Transformer architecture. Based on the unique routing mechanism, BRA and SSRA can calculate the scores between queries and keys and retain only top-k and top-m ones for each spatial and spectral token, through which the most irrelevant key–value pairs can be filtered out at a coarse-grained region level. After assuring the attention area, the token-to-token attention mechanism can be applied to extract global dependencies in both spatial and spectral domains. Additionally, to compensate for the local feature loss caused by BRA and SSRA, we develop a spatial–spectral position feedforward network (SSposFFN), which can enhance the nonlinear interrelationship and depth representations of features through convolution kernel and the ReLU. Finally, we designed a novel spatial–spectral dynamic fusion (SSDF) module to improve the discriminative capability of fusing cross-layer spatial–spectral feature representations.

In general, the main contribution of this work can be summarized as follows.

1) We developed a novel HSI classification network named SSFBF. Based on the "encoder–decoder" architecture, it can integrate global dependencies and local details in spatial and spectral domains.
2) We designed a dynamic routing attention mechanism in the spectral domain named SSRA to fully utilize the spectral bands of HSIs. Additionally, we developed an SSposFFN to replace the original MLP layer, thereby enhancing the nonlinear interrelationship representations from both spatial and spectral perspectives. Based on these modules, we improved the BiFormer Block and designed the SiFormer Block as the core modules for the spatial encoder and the spectral decoder, respectively.
3) We proposed an SSDF module called SSDF to achieve a multilevel cross-layer fusion between the spatial encoder and the spectral decoder, which can balance the feature representations between spatial and spectral branches and further improve the model's feature discrimination abilities.

The remainder of this paper is organized as follows. Section II introduces the details of the proposed SSFBF. Section III presents the dataset description, experimental settings, results, and discussion. Section IV provides the summary and conclusion.

II. Method

SSFBF consists of a spatial encoder, a spectral decoder, and SSDF modules. Fig. 1(a) shows the overall architecture of SSFBF.

Given an HSI patch input of fixed size $h \times w \times D$, shallow semantic information is initially extracted through a projecting operation via a CNN embedding module as $h \times w \times C$. Then, the transformed patch tokens are fed into a spatial encoder containing multiple BiFormer blocks and patch merging layers to generate hierarchical feature representations. Specifically, we use patch embedding for the first stage and patch merging for the second to fourth stages to reduce the spatial resolution and expand the channel dimension. We omitted the BiFormer block of stage 4 as the minor downsampling size results in fewer global dependencies. The original BiFormer block from [38] consists of a 3×3 depthwise convolution, a BRA module (see Section II-A), and a two-layer MLP with an expansion

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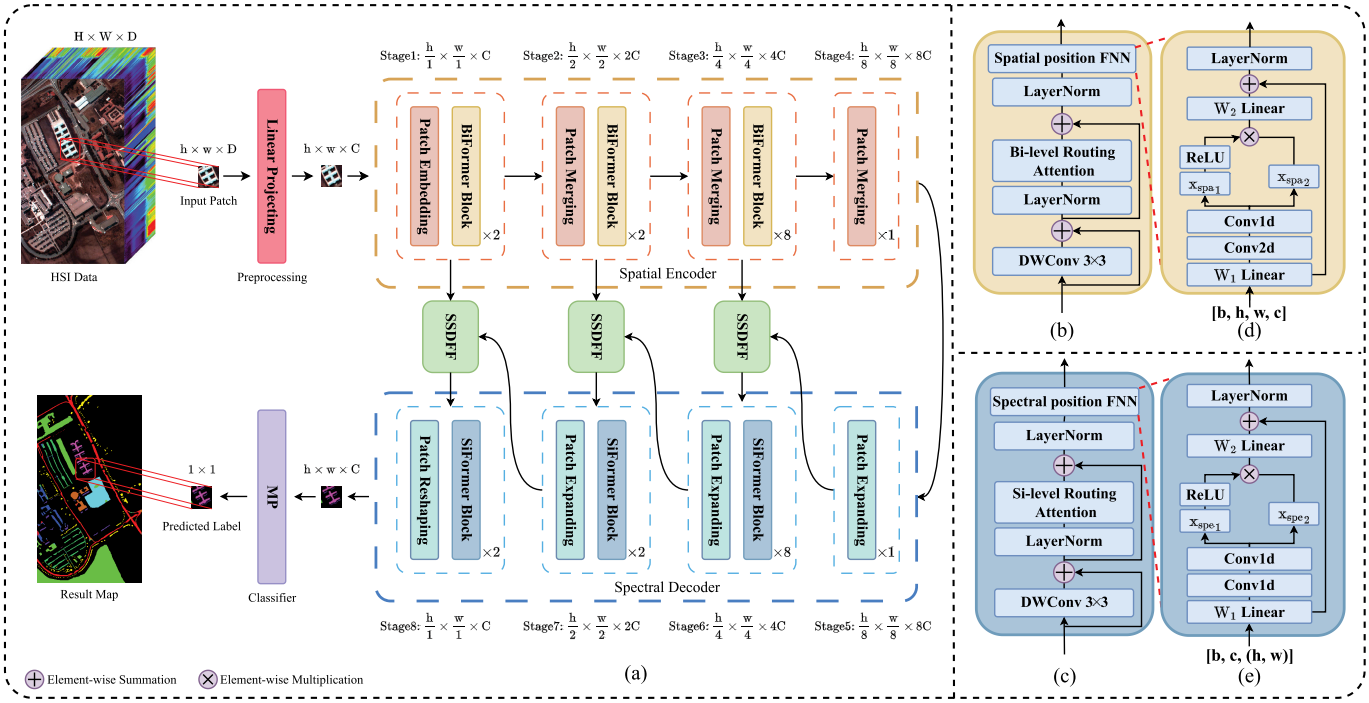


Fig. 1. (a) Overall architecture of SSFBF. A CNN module is used for channel projection. The spatial encoder employs BRA to capture spatial features. The spectral decoder then processes these feature maps via SSRA in the SiFormer block. Both BiFormer and SiFormer blocks incorporate an SSposFFN. Three SSDF modules facilitate multilevel cross-layer fusion between the encoder and the decoder, and an MLP predicts the pixel label from the high-level features. (b) BiFormer block. (c) SiFormer block. (d) Spatial position feedforward network. (e) Spectral position feedforward network.

rate of 3. Depth-wise convolution is used implicitly to encode relative position information, and two LayerNorm (LN) layers are employed to achieve normalization. Residual connections are utilized throughout the architecture. We improved the original BiFormer by replacing the MLP layer with the spatial position feedforward network proposed in Section II-C1, as shown in Fig. 1(b).

Subsequently, the output from the spatial encoder is fed into the spectral decoder. Inspired by [35], [39], we designed a spectral decoder, which consists of our designed SiFormer blocks and patch-expanding layers. The patch-expanding layers are used for spatial upsampling and channel reduction. We omitted the SiFormer block of stage 5 because the massive number of channels from downsampling leads to substantial computational overhead. As illustrated in Fig. 1(c), the SiFormer block initially employs a 3×3 depthwise convolution to implicitly encode relative positional information in channels, followed by SSRA block. The features are further enhanced through the spectral position feedforward network (Section II-C2), and the LN layers and residual connections are incorporated during the process.

The outputs of the SiFormer block are then upsampled and integrated with the spatial features from the Biformer into the SSDF for cross-layer spatial–spectral feature fusion. The fused features are fed into the next SiFormer block for further feature learning. Due to the omission of the BiFormer block and SiFormer block in stages 4 and 5, we utilize the results from patch merging and expanding as the input for the spectral decoder of the first SSDF module. The output from the final SiFormer block has just been reshaped to restore the original resolution of the feature map. These features are then fed into an MLP layer serving as the classification head, yielding the final results.

The procedure of SSFBF is presented in Algorithm 1. In the algorithm, encoder_stage_num and decoder_stage_num refer to the number of stages in the encoder and the decoder, respectively, while encoder_stage_block_num and decoder_stage_block_num denote the number of blocks in each stage. $\text{temx}_i^{\text{spa}}$ is used to store the output of each stage in the encoder temporarily.

A. Preliminaries: BRA

We use BRA as the core block of the spatial encoder. The BRA is a dynamic, query-ware sparse attention mechanism from [38]. It can filter out the most irrelevant key–value pairs at a coarse region level through a routing mechanism. Then, fine-grained token-to-token attention is used in these routed regions. BRA has a much lower complexity of $O((HW)^{(4/3)})$ compared with other traditional handcrafted static sparse attention mechanism [25], [40], [41] with $O((HW)^2)$.

1) Region Partition and Input Projection: Given an input patch of $\mathbf{X} \in \mathbb{R}^{h \times w \times C}$, it is divided into $S \times S$ nonoverlapped regions as $\mathbf{X}^r \in \mathbb{R}^{S^2 \times (hw/S^2) \times C}$. Then, the linear projections are used for deriving query, key, and value tensors, $\mathbf{Q}_{\text{spa}}, \mathbf{K}_{\text{spa}}, \mathbf{V}_{\text{spa}} \in \mathbb{R}^{S^2 \times (hw/S^2) \times C}$, as

$$\mathbf{Q}_{\text{spa}} = \mathbf{X}^r \mathbf{W}^q_{\text{spa}}, \quad \mathbf{K}_{\text{spa}} = \mathbf{X}^r \mathbf{W}^k_{\text{spa}}, \quad \mathbf{V}_{\text{spa}} = \mathbf{X}^r \mathbf{W}^v_{\text{spa}} \tag{1}$$

where $\mathbf{W}^q_{\text{spa}}, \mathbf{W}^k_{\text{spa}}, \mathbf{W}^v_{\text{spa}} \in \mathbb{R}^{C \times C}$ are projection weights for the query, key, and value tensors, respectively.

2) Region-to-Region Routing With the Directed Graph: First the region-level queries and keys, $\mathbf{Q}^r_{\text{spa}}, \mathbf{K}^r_{\text{spa}} \in \mathbb{R}^{S^2 \times C}$, are derived through per-region average. Then, matrix multiplication between $\mathbf{Q}^r_{\text{spa}}$ and $\mathbf{K}^r_{\text{spa}}$ is used to derive the adjacency

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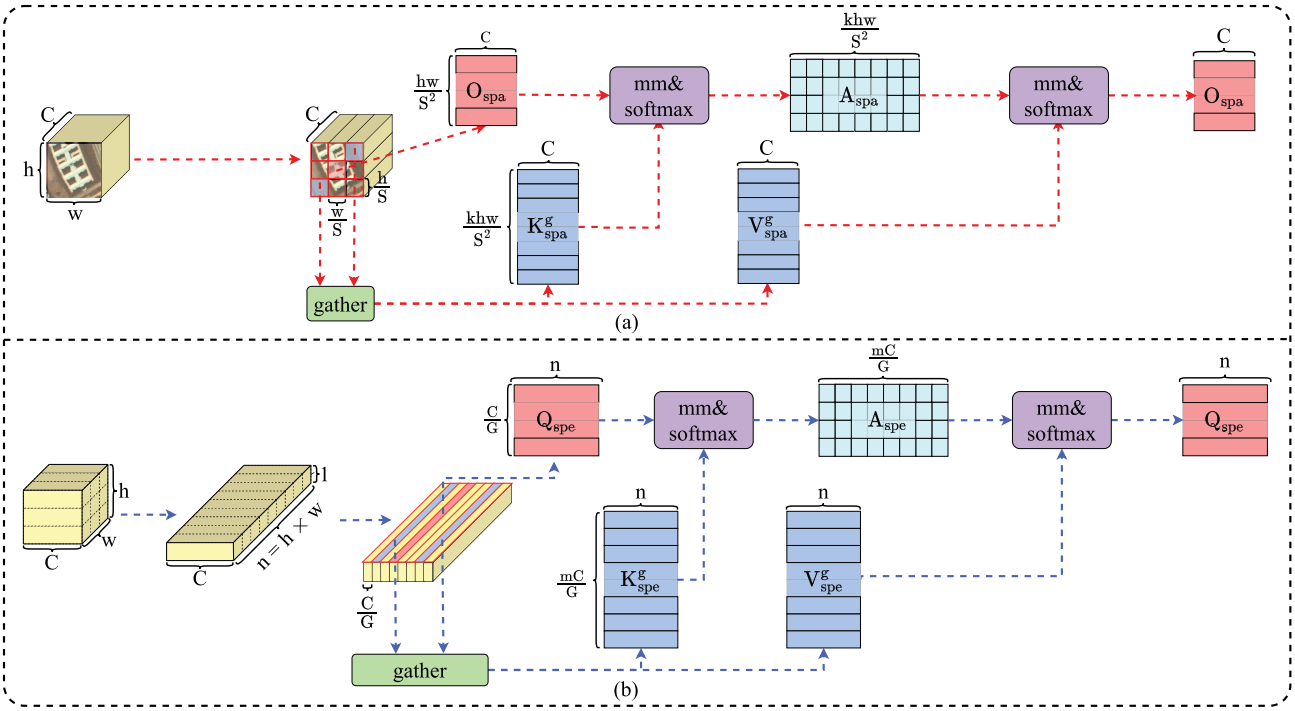


Fig. 2. (a) Illusion of BRA. Through gathering and calculating the most irrelevant spatial regions from the H and W dimensions, only GPU-friendly dense matrix multiplications can be involved. (b) Illusion of SSRA. A GPU-friendly spectral dimension attention mechanism is achieved by splitting consecutive spectral bands into segments and calculating the most relevant spectral areas from the channel dimension.

matrix of the region-to-region affinity graph. Subsequently, a routing index matrix, $\mathbf{I}^r_{\text{spa}} \in \mathbb{N}^{S^2\times k}$, is derived with a row-wise top-k operator to keep only top-k connections for each region of the graph

$$\mathbf{I}^r_{\text{spa}} = \text{topk Index}\left(\mathbf{Q}^r_{\text{spa}}\left(\mathbf{K}^r_{\text{spa}}\right)^T\right) \tag{2}$$

where topk Index$(\cdot)$ retains the indices of the k largest results from the multiplication, preserving the most relevant regions. Hence, the ith row of $\mathbf{I}^r_{\text{spa}}$ contains k indices of most relevant regions for the ith region.

3) Token-to-Token Attention: In the region-to-region routing index matrix $\mathbf{I}^r_{\text{spa}}$, fine-grained token-to-token attention is applied between each query token in region i and all key–value pairs residing in the union of k routed regions indexed with $\mathbf{I}^r_{\text{spa}_{(i,1)}}, \mathbf{I}^r_{\text{spa}_{(i,2)}}, \ldots, \mathbf{I}^r_{\text{spa}_{(i,k)}}$. Fig. 2(a) shows the process of the attention mechanism, as

$$\mathbf{K}^g_{\text{spa}} = \text{gather}\left(\mathbf{K}_{\text{spa}}, \mathbf{I}^r_{\text{spa}}\right) \tag{3}$$

$$\mathbf{V}^g_{\text{spa}} = \text{gather}\left(\mathbf{V}_{\text{spa}}, \mathbf{I}^r_{\text{spa}}\right) \tag{4}$$

$$\mathbf{O}_{\text{spa}} = \text{Attention}\left(\mathbf{Q}_{\text{spa}}, \mathbf{K}^g_{\text{spa}}, \mathbf{V}^g_{\text{spa}}\right) + \text{LCE}_{\text{spa}}(\mathbf{V}_{\text{spa}}) \tag{5}$$

where gather$(\cdot)$ is used to extract keys and values from $\mathbf{K}_{\text{spa}}$ and $\mathbf{V}_{\text{spa}}$ based on the index matrix $\mathbf{I}^r_{\text{spa}}$. $\mathbf{K}^g_{\text{spa}}, \mathbf{V}^g_{\text{spa}} \in \mathbb{R}^{S^2\times(khw/S^2)\times C}$ are gathered key and value tensors. $\mathbf{O}_{\text{spa}}$ is the output of the attention mechanism. $\text{LCE}_{\text{spa}}(\cdot)$ is a depth-wise convolution from [42] and the kernel size is 5.

B. Single-Level Spectral Routing Attention

In HSIs, each central pixel contains hundreds of channels with rich spectral information. We designed a dynamic routing attention mechanism called SSRA to fully utilize them and avoid the massive computational burden caused by numerous channels. SSRA can filter out the least relevant coarse-grained key–value pairs and retain the most relevant ones through the routing mechanism. We first split consecutive spectral bands into a specified number of segments. Then, we use the routing mechanism to choose the top-m channel segments with the strongest correlation. Finally, we perform the token-to-token attention operation on routed channel segments.

1) Channel Partitioning and Input Projection: Given an HSI input $\mathbf{X} \in \mathbb{R}^{h\times w\times C}$, we first transpose it to obtain $\mathbf{X}^T \in \mathbb{R}^{C\times h\times w}$ and flatten h and w into a token sequence n, as $\mathbf{X}^T \in \mathbb{R}^{C\times n}$. We divide it into G nonoverlapped spectral segments as $\mathbf{X}^{\mathbf{T}r} \in \mathbb{R}^{G\times(C/G)\times n}$. Then, we derive the query, key, and value tensor, Q_{spe}, K_{spe}, $V_{\text{spe}} \in \mathbb{R}^{G\times(C/G)\times n}$, by linear projections

$$\mathbf{Q}_{\text{spe}} = \mathbf{X}^{\mathbf{T}r}\mathbf{W}^q_{\text{spe}}, \quad \mathbf{K}_{\text{spe}} = \mathbf{X}^{\mathbf{T}r}\mathbf{W}^k_{\text{spe}}, \quad \mathbf{V}_{\text{sqe}} = \mathbf{X}^{\mathbf{T}r}\mathbf{W}^v_{\text{spe}} \tag{6}$$

where $\mathbf{W}^q_{\text{spe}}$, $\mathbf{W}^k_{\text{spe}}$, and $\mathbf{W}^v_{\text{spe}} \in \mathbb{R}^N$ are projection weights for the query, key, and value along token sequence dimension, respectively.

2) Spectral Segment-to-Segment Routing Mechanism: We first derive the affinity between fragments by calculating the segment-based queries and keys. Then, the critical routing

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Algorithm 1 Procedure of SSFBF

Input: The input HSI data $\mathbf{I} \in \mathbb{R}^{H\times W\times D}$;
Output: The predicted labels $\mathbf{Y}$.
Forward:
for one cropped patch $x_o \in \mathbb{R}^{h\times w\times D}$ from $\mathbf{I}$ **do**
  CNN embedding on x_o to generate $x \in \mathbb{R}^{h\times w\times C}$.
  for $i = 0$ **to** encoder_stage_num-1 **do**
    if $i = 0$ **then**
      x_i = Patch Embedding(x)
    else
      x_i = Patch Merging(x_{i-1})
    end if
    if i $\neq$ encoder_stage_num-1 **do**
      for $j = 0$ **to** encoder_stage_block_num-1 **do**
        $x_i = x_i +$ pos_embed(x_i)
        $x_i = x_i +$ BRA(x_i)
        $x_i = x_i +$ S_{spa}posFFN(x_i)
      end for
    end if
    $temx_i^{spa} = x_i$
  end for
  for i = decoder_stage_num-1 **to** 0 **do**
    if i $\neq$ decoder_stage_num-1 **then**
      x_i = SSDF($temx_i^{spa}$, x_{i+1})
      for $j = 0$ **to** decoder_stage_block_num-1 **do**
        $x_i = x_i +$ pos_embed(x_i)
        $x_i = x_i +$ SSRA(x_i)
        $x_i = x_i +$ S_{spe}posFFN(x_i)
      end for
    end if
    if i $\neq$ 0 **do**
      x_i = Patch Expanding(x_i)
    else
      x = Reshape(x_i)
    end if
  $\mathbf{y}$ = MLP(x)
end for
Predict the corresponding labels of $\mathbf{y} \in \mathbf{Y}$

mechanism is used to retain the most relevant spectral areas for each query segment. Specifically, we derive the sublevel queries and keys, $\mathbf{Q}^r_{\text{spe}}, \mathbf{K}^r_{\text{spe}} \in \mathbb{R}^{G\times N}$, by calculating the average of each spectral segment of $\mathbf{Q}_{\text{spe}}$ and $\mathbf{K}_{\text{spe}}$. Then, we derive the adjacency matrix between fragments through matrix multiplication of $\mathbf{Q}^r_{\text{spe}}$ and transposed $\mathbf{K}^r_{\text{spe}}$. Finally, the top-$m$ operator is performed to retain the most relevant m areas for each query segment, resulting in the core routing index matrix $\mathbf{I}^r_{\text{spe}} \in \mathbb{N}^{G\times m}$

$$\mathbf{I}^r_{\text{spe}} = \text{topm Index}\left(\mathbf{Q}^r_{\text{spe}}\left(\mathbf{K}^r_{\text{spe}}\right)^T\right) \tag{7}$$

where topm Index$(\cdot)$ extracts the indices of the top m maximum values of matrix multiplication. Hence, the ith row of $\mathbf{I}^r_{\text{spe}}$ contains m indices of most relevant areas for the ith segment.

3) Token-to-Token Spectral Attention Mechanism: We develop a token-based spectral attention mechanism to process the adjacency matrix of spectral segments above. In spectral segment j, each query token will attend to all key-value pairs within the most relevant m segments, denoted as $I_{\text{sqe}(j,1)}$, $I_{\text{sqe}(j,2)}, \ldots, I_{\text{sqe}(j,m)}$. However, these segments may be scattered throughout the whole band range. Thus, we gather key and value tensors and calculate the spectral attention for the gathered key–value pairs as follows:

$$\mathbf{K}^g_{\text{spe}} = \text{gather}_{\text{spe}}\left(\mathbf{K}_{\text{spe}}, \mathbf{I}^r_{\text{spe}}\right) \tag{8}$$

$$\mathbf{V}^g_{\text{spe}} = \text{gather}_{\text{spe}}\left(\mathbf{V}_{\text{spe}}, \mathbf{I}^r_{\text{spe}}\right) \tag{9}$$

$$\mathbf{O}_{\text{spe}} = \text{Attention}\left(\mathbf{Q}_{\text{spe}}, \mathbf{K}^g_{\text{spe}}, \mathbf{V}^g_{\text{spe}}\right) + \text{LCE}_{\text{spe}}(\mathbf{V}_{\text{spe}}) \tag{10}$$

where gather$(\cdot)$ collects keys and values in $\mathbf{K}_{\text{spe}}$ and $\mathbf{V}_{\text{spe}}$ according to the index matrix $\mathbf{I}^r_{\text{spe}}$. $\mathbf{K}^g_{\text{spe}}, \mathbf{V}^g_{\text{spe}} \in \mathbb{R}^{G\times(mC/G)\times N}$ are the gathered key and value tensors. $\mathbf{O}_{\text{spe}}$ is the output of the attention mechanism. Additionally, we add a parallel local spectral enhancement term as $\text{LCE}_{\text{spe}}(\cdot)$, which can extract local fine-grained features between bands to complement the global feature extraction of the attention mechanism. We set the kernel size to 5. The whole process is shown in Fig. 2(b).

C. Spatial–Spectral Position Feedforward Network

Inspired by [43], we developed a novel SSposFFN based on pointwise convolutions and the ReLU. It replaces the original MLP layer in the BiFormer and SiFormer block, further addressing the local feature loss from the attention mechanism. We introduced different position feedforward networks for BRA and SSRA. Fig. 1(d) and (e) displays the architecture of spatial and spectral position feedforward networks, respectively.

1) Spatial Position Feedforward Network: Given an input $x_{\text{spa}} \in \mathbb{R}^{h\times w\times d}$ from BRA, we first adjust the channels to d_{hid} through a linear projection layer. Then, we double the number of channels through a 2-D CNN and further preserve the dimensionality through a 1-D CNN with a kernel size of 1. We divide it into two interactive features along the channel dimension as

$$x'_{\text{spa}} = \text{Cov}^{\text{up}}_{2d}\left(W_1 x_{\text{spa}}\right) \tag{11}$$

$$x_{\text{spa}_1}, x_{\text{spa}_2} = \text{Split}\left(\text{Cov1d}\left(x'_{\text{spa}}\right)\right) \tag{12}$$

where $W_1 \in \mathbb{R}^{h\times w\times \text{hid}}$ represents the weight matrix of the linear projection layer and $\text{Cov}^{\text{up}}_{2\text{d}}$ refers to the upsampling 2-D CNN for h and w that doubles the channel dimensions as $x'_{\text{spa}} \in \mathbb{R}^{h\times w\times 2d_{\text{hid}}}$. Cov1d is used to further maintain features unchanged, and $x_{\text{spa}_1}, x_{\text{spa}_2} \in \mathbb{R}^{h\times w\times d_{\text{hid}}}$ signify the two interactive features separated from $x_{\text{spa}}{}'$.

Then, we use the ReLU to process x_{spa_1} and perform elementwise multiplication with x_{spa_2}. Finally, we recover the dimension and add a residual connection, as follows:

$$x''_{\text{spa}} = \text{ReLU}\left(x_{\text{spa}_1}\right) * x_{\text{spa}_2} \tag{13}$$

$$o_{\text{spa}} = W_2\left(x''_{\text{spa}}\right) + x_{\text{spa}} \tag{14}$$

where $x''_{\text{spa}} \in \mathbb{R}^{h\times w\times d_{\text{hid}}}$ signifies the results of elementwise multiplication. o_{spa} represents the total outputs. $W_2 \in \mathbb{R}^{h\times w\times d}$ refers to the weight matrix of the linear projection layer for recovering.

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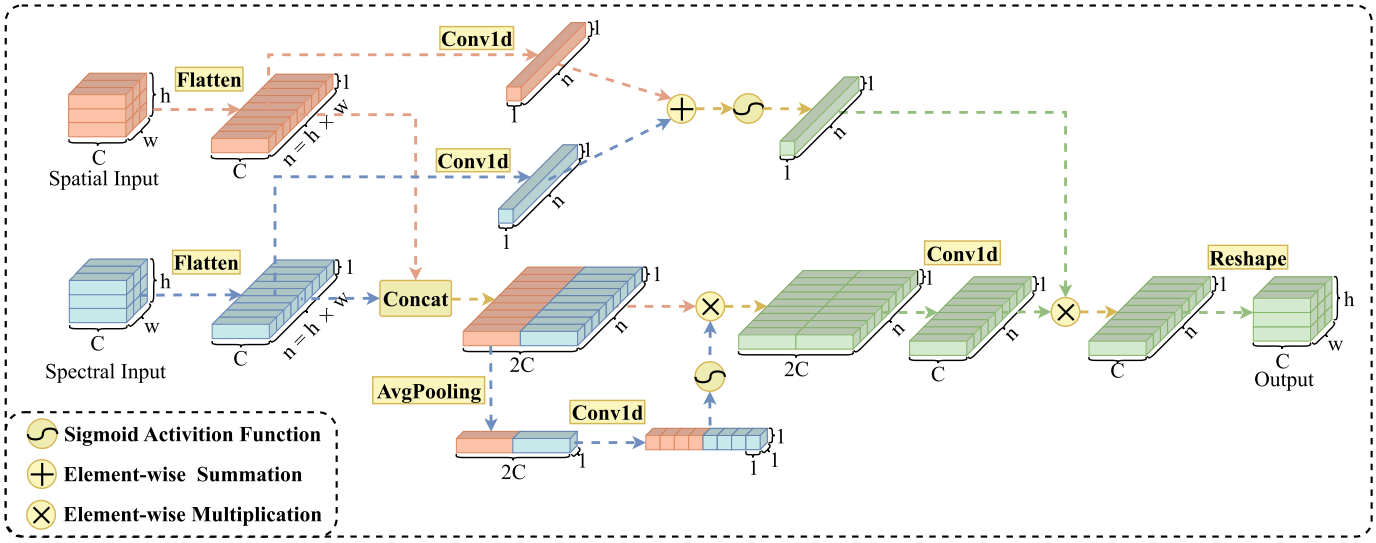


Fig. 3. Visualization process of SSDF. Global and local branches are used to fuse spatial–spectral features. The global branch assigns each position a global weight through the adaptive average pooling and then applies it to concatenated features. The local branch emphasizes the local weight of each position through 1-D CNNs and summation and applies it to the integrated global feature map. This dual-branch architecture can dynamically fuse global and local spatial–spectral representations.

2) Spectral Position Feedforward Network: We develop a spectral position feedforward network to enhance the discrimination of the spectral representation from SSRA. Specifically, given an input $x_{\text{spe}} \in \mathbb{R}^{h\times w\times d}$ from SSRA, we first flatten the h and w into the token sequence n and transpose the feature, as $x_{\text{spe}} \in \mathbb{R}^{d\times n}$, and adjust it to n_{hid}. Then, we use a 1-D CNN to expand the token to twice its original size and further preserve the dimension through another 1-D CNN with a kernel size of 1. Subsequently, we divide it into two interactive features along the token dimension, as follows:

$$x'_{\text{spe}} = \text{Cov}_{1d}^{\text{up}}\left(W_1 x_{\text{spe}}\right) \tag{15}$$

$$x_{\text{spe}_1}, x_{\text{spe}_2} = \text{Split}\left(\text{Cov1d}\left(x'_{\text{spe}}\right)\right) \tag{16}$$

where $W_1 \in \mathbb{R}^{d\times n_{\text{hid}}}$ represents the weight matrix of linear projection layer. $\text{Cov}_{1\text{d}}^{\text{up}}$ refers to the upsampling 1-D CNN for d that doubles the token dimensions as $x'_{\text{spe}} \in \mathbb{R}^{d\times 2n_{\text{hid}}}$. Cov1d is used to further maintain the dimensions, and $x_{\text{spe}_1}, x_{\text{spe}_2} \in \mathbb{R}^{d\times n_{\text{hid}}}$ signify the two interactive features separated from x'_{spe}.

Then, x_{spe_1} is processed by ReLU and made elementwise multiplication with x_{spe_2}. Finally, we recover the dimension, and a residual connection is added, as follows:

$$x''_{\text{spe}} = \text{ReLU}\left(x_{\text{spe}_1}\right) * x_{\text{spe}_2} \tag{17}$$

$$o_{\text{spe}} = W_2\left(x''_{\text{spe}}\right) + x_{\text{spe}} \tag{18}$$

where $x''_{\text{spe}} \in \mathbb{R}^{d\times d_{\text{hid}}}$ signifies the results of elementwise multiplication. o_{spe} represents the total outputs of spectral position feedforward network. $W_2 \in \mathbb{R}^{d\times n}$ refers to the weight matrix of the linear projection layer for recovering.

D. Spatial–Spectral Dynamic Fusion

Inspired by [44], we designed an SSDF mechanism between the BiFormer and SiFormer blocks. SSDF can enhance the dimensional interaction of multiscale features through cross-layer fusion, thereby achieving a further output of fine-grained spatial–spectral details. Specifically, we integrate the spatial and spectral inputs by employing two dynamic selection branches: global context extraction and local details extraction branches. Fig. 3 illustrates the visualization process of SSDF.

Given the spatial input $x_{\text{spa}} \in \mathbb{R}^{h\times w\times c}$ and spectral input $x_{\text{spe}} \in \mathbb{R}^{h\times w\times c}$, we first flatten h and w into token sequence n, as $x_{\text{spa}}, x_{\text{spe}} \in \mathbb{R}^{n\times c}$. For the global context branch, we concatenate x_{spa} and x_{spe} along the channel dimension and apply an adaptive average pooling layer to the concatenated feature maps, compressing the channel dimension to 1 to extract global features. Furthermore, a 1-D CNN with a kernel size of 1 and sigmoid are used to obtain the weight matrix. The weight matrix is applied to the concatenated feature by elementwise multiplication, assigning each position a weight derived from the global context. Finally, we use a 1-D CNN for dimension recovery, as follows:

$$W_{\text{glo}} = \text{Sigmoid}\left(\text{Conv1d}\left(\text{AVGPool}\left(\left[x_{\text{spa}}; x_{\text{spe}}\right]_c\right)\right)\right) \tag{19}$$

$$x_{\text{glo}} = \text{Conv1d}\left(\text{W}_{\text{glo}} \otimes x\right) \tag{20}$$

where x_{glo} is the output of global context branch and W_{glo} refers to the weight matrix. $x_{\text{spa}}, x_{\text{spe}} \in \mathbb{R}^{n\times c}$ are the spatial and spectral inputs, concatenated along the channel dimension.

For the local detailed feature extraction branch, we first obtain the feature of each position by processing x_{spa} and x_{spe} with two 1-D CNNs with a kernel size of 1. The results are then summed and fused, followed by applying sigmoid to generate a local weight matrix, as follows:

$$\text{W}_{\text{loc}} = \text{Sigmoid}\left(\text{Conv1d}\left(x_{\text{spa}}\right) \oplus \text{Conv1d}\left(x_{\text{spe}}\right)\right) \tag{21}$$

where W_{loc} refers to the weight matrix output of the detail feature extraction branch.

Finally, we apply the local weight matrix to the global output. The weights are broadcasted into the output through elementwise multiplication, ensuring that the output features

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TABLE I
CLASS DISTRIBUTION AND SAMPLE COUNTS FOR TRAINING, VALIDATING, AND TESTING ACROSS FOUR DATASETS

No.	Indian Pines (IP)				Salinas Scene (SA)				Houston 2013 (HU)				Kennedy Space Center (KSC)			
	Class	Train.	Validate.	Test	Class	Train.	Validate.	Test	Class	Train.	Validate.	Test	Class	Train.	Validate.	Test
1	Alfalfa	2	5	39	Brocoli green weeds1	20	201	1788	Healthy grass	63	125	1063	Scrub	76	76	609
2	Corn-notill	71	143	1214	Brocoli green weeds2	37	372	3317	Stressed grass	63	125	1066	Willow swamp	24	24	195
3	Corn-mintill	41	83	706	Fallow	21	210	1868	Synthetic grass	35	70	592	CP hammock	26	26	204
4	Corn	12	24	201	Fallow rough plow	20	139	1241	Trees	62	124	1058	Slash pine	25	25	202
5	Grass-pasture	24	48	411	Fallow smooth	27	268	2383	Soil	62	124	1056	Oak/Broadleaf	16	16	129
6	Grass-trees	37	73	620	Stubble	39	396	3524	Water	16	33	276	Hardwood	23	23	183
7	Grass-pasture-mowed	1	3	24	Celery	36	358	3185	Residential	63	127	1078	Swap	11	11	83
8	Hay-windrowed	24	48	406	Grapes untrained	113	1127	10031	Commercial	62	124	1058	Graminoid marsh	43	43	345
9	Oats	1	2	17	Soil vineyard develop	62	620	5521	Road	63	125	1064	Spartina marsh	52	52	416
10	Soybean-notill	49	97	826	Corn senesced green weeds	33	328	2917	Highway	61	123	1043	Cattail marsh	40	40	324
11	Soybean-mintill	123	245	2087	Lettuce romaine 4wk	11	107	950	Railway	62	123	1050	Salt marsh	42	42	335
12	Soybean-clean	30	59	504	Lettuce romaine 5wk	19	193	1715	Parking Lot 1	62	123	1048	Mud flats	50	50	403
13	Wheat	10	20	175	Lettuce romaine 6wk	9	91	816	Parking Lot 2	23	47	399	Water	93	93	741
14	Woods	63	126	1076	Lettuce romaine 7wk	11	107	952	Tennis Court	21	43	364				
15	Buildings-Grass-Trees-Drives	19	39	328	Vinyard untrained	72	727	6469	Running Track	33	66	561				
16	Stone-Steel-Towers	5	9	79	Vinyard vertical trellis	18	181	1608								
	total	512	1024	8713	total	541	5412	48176	total	751	1502	12776	total	521	521	4169

encompass global contextual information and local fine-grained details. Furthermore, we produce the output as follows:

$$x_{\text{out}} = \text{W}_{\text{loc}} \otimes x_{\text{glo}} \tag{22}$$

where x_{out} is the fused output of SSDF.

E. Loss Function

We employed the cross-entropy loss function during the training process. The cross-entropy loss function is defined as follows:

$$L_{\text{ce}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{C} G(k,i)\cdot\log(P(k,i)) \tag{23}$$

where N is the number of pixels and $G(k,i) \in (0,1)$ and $P(k,i) \in (0,1)$ indicates the ground truth and the probability of the ith pixel for class k, respectively.

III. EXPERIMENTS

A. Data Description

We selected four datasets under varying conditions to validate the performance of our SSFBF. The details for all datasets are presented in Table I.

1) Indian Pines: The Indian Pines (IP) dataset was collected by an AVIRIS sensor in northwestern India. The dataset contains 145 × 145 pixels and 224 spectral bands, with 16 land-cover classes. The bands are reduced to 200 by removing 24 water-absorption and noise bands. We randomly select 5%, 10%, and 85% for training, validation, and testing, respectively.

2) Salinas Scene: The Salinas Scene (SA) dataset is captured by an AVIRIS sensor in Salinas Valley, California. The dataset contains 512 × 217 pixels and 224 spectral bands, including 16 classes. We adopt 204 bands for experiments and randomly selected 1%, 10%, and 89% for training, validation, and testing, respectively.

3) Houston 2013: The Houston 2013 (HU) dataset [45] was acquired by the NCALM at the University of Houston. It was first used in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest. The dataset comprises 249 × 1905 pixels and 144 bands, covering nine land-cover categories. We randomly select 5%, 10%, and 85% for training, validation, and testing, respectively.

4) Kennedy Space Center: The Kennedy Space Center (KSC) dataset was collected by the AVIRIS sensor over the Kennedy Space Center in Florida. The dataset consists of 512 × 614 pixels and 224 bands. 13 classes are defined, and 176 bands are used for the analysis. We randomly select 10%, 10%, and 80% for training, validation, and testing, respectively.

B. Experimental Setting

1) Evaluation Metrics: We adopted overall accuracy (OA), average accuracy (AA), Kappa coefficient (k), and the classification accuracy of each land-cover category as indicators for quantitative analysis.

2) Comparison Methods: The proposed method is compared with several state-of-the-art methods, including HybridSN [18], DBDA [46], Hit [33], MFT [47], SS1DSwin [28], MF [36], 3D-SRT [31], CLOLN [23], and MASSF [37]. HybridSN, DBDA, and CLOLN are based on CNN, while MFT, SS1DSwin, and 3D-SRT are recently proposed Transformer-based methods. Hit, MF, and MASSF are methods combining CNN and Transformer modules.

For our SSFBF, we set the initial patch size as 16 × 16 for all datasets. The depth of the spatial encoder was [2, 2, 8, 1], and the spectral decoder's depth was [2, 8, 2, 1]. Moreover, the drop rate was set as 0.2. The region partition factor for the BRA was set as 8. The head dim was set as 32. The number of top-k was set as [2, 4, 8, −2]. For the SSRA, the band partition factor splitting the channels was set as 16. The head dim was set as 8. And the number of top-m was set as [2, 4, 8, −2]. A value of −2 in top-k and top-m indicates that neither BRA nor SSRA are utilized in this stage.

3) Implementation Details: We used the Adam optimizer with an initial learning rate of 0.0001 and a weight decay factor of 1×10^{-6} for L2 regularization in the training of our SSFBF and other comparison methods. The batch size was set to 128. The training epoch was set to 300, with validation performed every 5 epoch. The best model determined by validation was subsequently adopted for testing. All experiments were conducted ten times, and the average result was recorded. All the methods are implemented on the Pytorch platform. The device was an Intel Core i5-13490F CPU and Nvidia RTX 3090 24G GPU.

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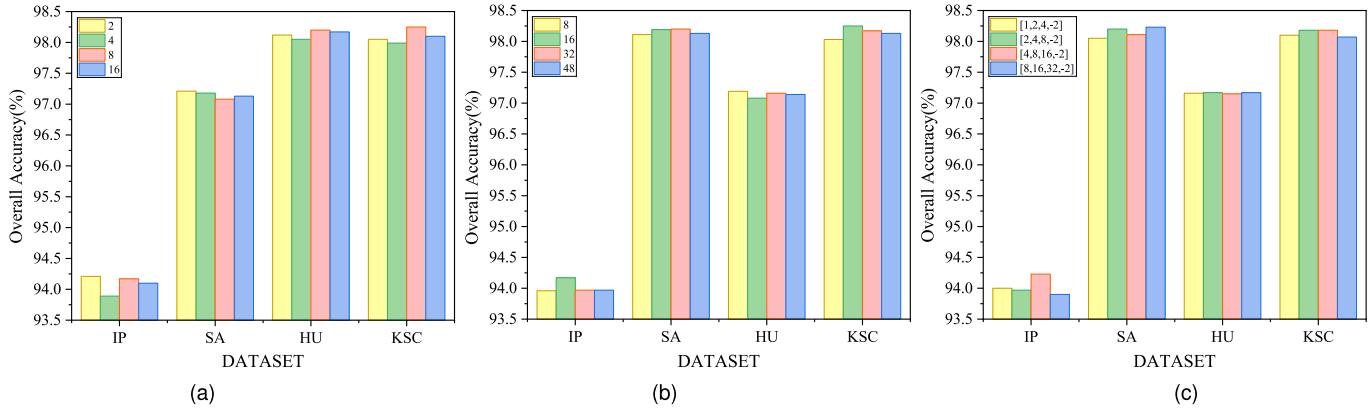


Fig. 4. Performance of SSFBF on four datasets with different numbers of heads, spectral band partition factors, and spectral top-m. (a) Effects of the number of heads in SSRA. (b) Effects of spectral band partition factor. (c) Effects of spectral top-m.

C. Parameter Analysis

To develop a comprehensive study of SSFBF, we analyzed the number of attention heads in SSRA, the partition size of subspectral segments, and the optimal number of preserved indices in SSRA as top-m. These hyperparameters significantly influenced the overall performance of the model. We configured other parameters as presented in the experimental Section III-B2.

1) Effects of the Number of Heads in SSRA: To determine the optimal number of attention heads in SSRA, 2, 4, 8, and 16 were chosen. We fixed the spectral partition size as 32 and the number of spectral top-m as $[2, 4, 8, -2]$. As shown in Fig. 4(a), the best number is 8 for IP and SA and 16 for HU and KSC. Considering all the datasets, we choose 8 as the optimal number of attention heads in SSRA.

2) Effects of the Spectral Band Partition Factor: Additionally, we evaluate the effect of the spectral band partition factor on classification accuracy, which indicates the number of spectral segments split from consecutive spectral bands. The size of partition factors determines how much spectral band information each spectral segment contains, which is selected from 8, 16, 32, and 48. We fixed the number of heads in SSRA as 8 and the number of spectral top-m as $[2, 4, 8, -2]$. As shown in Fig. 4(b), the best factor is 16 for IP and KSC, 8 for HU, and 32 for SA. It can be seen that increasing the spectral band partition factor does not necessarily result in the acquisition of more spectral band information for each segment. It can be attributed to each segment's noise and the imbalanced distribution of influential spectral bands in the channel dimension. As a result, we chose 16 as the optimal partition factor for all datasets.

3) Effects of Spectral Top-m: To verify the effectiveness of the routing index strategy in spectral bands, we test the effect of spectral top-m. The spectral top-m determines the number of most relevant m segments that should be retained, which is selected from $[1, 2, 4, -2]$, $[2, 4, 8, -2]$, $[4, 8, 16, -2]$ and $[8, 16, 32, -2]$. We fixed the number of heads in SSRA as 8 and the spectral partition size as 32. The results are shown in Fig. 4(c). For the IP and SA, the best top-m is $[4, 8, 16, -2]$ and $[8, 16, 32, -2]$. $[2, 4, 8, -2]$ and $[8, 16, 32, -2]$ are both optimal for the HU dataset. $[2, 4, 8, -2]$ and $[4, 8, 16, -2]$ are the best options for the KSC dataset. It is worth mentioning that the accuracy does not consistently increase with the number of retained segments for all datasets. This could be explained by the excessive number of retained segments providing redundant band information, adversely affecting the performance. Conversely, retaining too few indexed segments fails to extract sufficient spectral information from context, leading to a decline in performance. However, it can be seen that $[2, 4, 8, -2]$ achieved overall notable performance for all datasets. Therefore, we chose it as the optimal spectral top-m.

TABLE II

ABLATION STUDY RESULTS OF MAIN MODULES ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Classes	Module			Datasets			
	spectral decoder	SSposFFN	SSDF	IP	SA	HU	KSC
1	✗	✗	✗	85.85	93.87	94.05	89.22
2	✓	✗	✗	85.68	95.76	94.39	91.48
3	✗	✓	✗	90.23	93.76	95.73	92.64
4	✗	✗	✓	89.02	97.29	95.57	96.56
5	✓	✓	✗	91.43	95.63	95.99	93.71
6	✓	✗	✓	89.02	97.29	95.57	96.56
7	✓	✓	✓	**94.17**	**98.19**	**97.08**	**98.25**

TABLE III

ABLATION STUDY RESULTS OF THE SPECTRAL DECODER ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Module					Dataset			
Spatial Encoder (BiFormer)	Spectral Decoder		SSposFFN	SSDF	IP	SA	HU	KSC
	BiFormer	SiFormer						
✓	✗	✗	✗	✗	85.52	93.87	94.05	89.22
✓	✓	✗	✗	✗	**85.87**	**95.87**	94.28	90.60
	✗	✓			85.68	95.76	**94.39**	**91.48**
✓	✓	✗	✓	✗	88.06	95.28	94.14	93.04
	✗	✓			**91.43**	**95.63**	**95.99**	**93.71**
✓	✓	✗	✗	✓	88.71	97.09	95.24	96.15
	✗	✓			**89.02**	**97.29**	**95.57**	**96.56**
✓	✓	✗	✓	✓	92.54	97.88	96.58	97.87
	✗	✓			**94.17**	**98.19**	**97.08**	**98.25**

TABLE IV

ABLATION STUDY RESULTS OF THE SSPOSFFN ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Classes	Module				Datasets			
	Spectral decoder	SSDF	PosFFN	SSposFFN (ours)	IP	SA	HU	KSC
1(baseline)	✗	✗	✗	✗	85.52	93.87	94.05	89.22
2	✗	✗	✓	✗	88.77	**95.18**	95.49	92.39
3	✗	✗	✗	✓	**90.23**	93.76	**95.73**	**92.64**
4	✓	✗	✓	✗	88.25	95.81	94.87	92.88
5	✓	✗	✗	✓	**89.02**	**97.29**	**95.57**	**96.56**
6	✓	✓	✓	✗	92.55	98.08	96.87	97.96
7	✓	✓	✗	✓	**94.17**	**98.19**	**97.08**	**98.25**

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TABLE V
CLASSIFICATION RESULTS OF THE DIFFERENT METHODS ON THE INDIA PINES DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Class	CNN-based Methods			Transformer-based Methods			CNN-Transformer-based Methods			
	HybridSN	DBDA	CLOLN	MFT	SS1DSwin	3D-SRT	Hit	MF	MassF	SSFBF
Alfalfa	95.60 (±4.35)	00.00 (±0.00)	84.71 (±23.93)	93.67 (±9.45)	79.60 (±15.80)	73.14 (±18.93)	94.81 (±4.59)	**99.27 (±1.46)**	90.57 (±10.88)	98.65 (±3.12)
Corn-notill	79.34 (±3.08)	82.21 (±5.34)	**93.02 (±6.24)**	88.19 (±4.72)	75.57 (±3.06)	68.65 (±5.18)	66.55 (±5.46)	86.27 (±3.84)	89.86 (±1.84)	88.78 (±1.89)
Corn-mintill	87.52 (±2.08)	79.94 (±8.56)	90.39 (±7.31)	84.18 (±5.54)	74.82 (±3.88)	68.21 (±9.21)	72.68 (±8.64)	92.09 (±2.29)	89.72 (±1.91)	**94.42 (±2.72)**
Corn	86.80 (±5.35)	**98.71 (±2.07)**	95.09 (±6.91)	89.85 (±3.93)	81.06 (±6.24)	59.95 (±10.62)	80.71 (±6.22)	91.84 (±5.57)	93.26 (±2.59)	96.31 (±1.73)
Grass-pasture	94.19 (±3.04)	94.33 (±2.43)	**94.49 (±3.57)**	94.09 (±5.28)	90.05 (±3.69)	91.30 (±4.43)	90.32 (±2.95)	94.35 (±1.49)	91.64 (±3.84)	93.80 (±1.61)
Grass-trees	94.89 (±2.60)	92.84 (±5.41)	93.78 (±6.22)	96.18 (±1.31)	87.61 (±4.57)	89.21 (±4.80)	91.04 (±2.84)	95.43 (±1.09)	92.93 (±1.85)	**96.67 (±0.85)**
Grass-pasture-mowed	98.57 (±2.86)	00.00 (±0.00)	59.09 (±34.24)	71.10 (±34.18)	85.32 (±11.06)	48.17 (±30.23)	80.00 (±40.00)	92.95 (±6.70)	74.42 (±39.12)	**98.82 (±3.53)**
Hay-windrowed	95.36 (±1.33)	91.58 (±1.24)	**99.44 (±1.53)**	96.58 (±1.88)	94.25 (±1.37)	89.83 (±2.86)	96.47 (±1.92)	98.38 (±0.96)	92.51 (±2.34)	98.89 (±1.00)
Oats	86.50 (±22.14)	00.00 (±0.00)	67.75 (±20.77)	42.62 (±39.52)	36.67 (±33.47)	71.69 (±30.17)	55.00 (±41.53)	86.33 (±14.16)	60.00 (±48.99)	**99.00 (±3.00)**
Soybean-notill	90.99 (±2.69)	88.79 (±7.71)	93.41 (±4.77)	90.49 (±4.50)	74.33 (±2.58)	76.37 (±7.82)	80.47 (±3.19)	92.30 (±1.90)	94.25 (±1.89)	**94.26 (±2.70)**
Soybean-mintill	92.82 (±1.19)	89.59 (±3.85)	91.30 (±4.34)	92.77 (±3.24)	85.65 (±3.56)	75.29 (±5.69)	83.08 (±3.50)	92.14 (±1.70)	93.32 (±1.40)	**95.19 (±0.48)**
Soybean-clean	78.00 (±5.93)	78.95 (±5.56)	**88.93 (±6.48)**	85.60 (±3.40)	56.96 (±5.36)	58.99 (±5.96)	62.49 (±6.21)	79.82 (±3.54)	86.38 (±3.81)	82.78 (±3.97)
Wheat	98.02 (±1.73)	99.41 (±0.73)	95.86 (±3.55)	97.42 (±4.17)	93.98 (±2.97)	92.01 (±5.76)	99.44 (±0.50)	98.92 (±1.67)	98.65 (±1.60)	**99.83 (±0.26)**
Woods	95.23 (±1.35)	91.71 (±2.03)	98.48 (±0.69)	97.12 (±1.82)	94.63 (±1.03)	91.77 (±2.07)	96.14 (±0.95)	97.60 (±1.16)	96.63 (±1.41)	**98.81 (±0.34)**
Buildings-Grass-Trees-Drives	87.90 (±2.98)	89.76 (±4.37)	91.58 (±6.18)	86.75 (±6.30)	79.63 (±2.55)	70.99 (±7.13)	88.76 (±5.01)	87.65 (±3.90)	92.57 (±2.83)	**96.81 (±1.64)**
Stone-Steel-Towers	91.03 (±3.76)	98.19 (±1.97)	89.93 (±3.93)	81.44 (±8.54)	93.43 (±4.95)	88.95 (±5.96)	83.84 (±2.71)	**94.90 (±3.50)**	87.73 (±4.78)	94.78 (±2.95)
OA	89.60 (±0.59)	87.69 (±1.50)	92.49 (±1.50)	91.23 (±1.71)	82.12 (±1.84)	77.13 (±3.68)	81.82 (±2.05)	91.74 (±1.05)	92.42 (±0.86)	**94.17 (±0.68)**
AA	79.88 (±1.68)	69.40 (±2.13)	**87.29 (±4.44)**	81.31 (±3.42)	72.63 (±2.61)	68.16 (±4.30)	70.33 (±3.41)	84.20 (±1.68)	78.70 (±2.98)	84.57 (±2.21)
k×100	88.13 (±0.67)	85.90 (±1.72)	91.42 (±1.72)	90.00 (±1.95)	79.59 (±2.14)	73.84 (±4.22)	79.21 (±2.38)	90.57 (±1.20)	91.34 (±0.98)	**93.34 (±0.78)**

D. Ablation Studies

To evaluate the contribution of each module in our proposed SSFBF model, we designed comprehensive ablation studies with different combinations of modules on four datasets. The complete SSFBF model comprises four main modules: the spatial encoder, the spectral decoder, SSposFFN, and SSDF. In addition, we further tested the effect of the spectral decoder and SSposFFN. The experimental details were as follows.

1) Ablation Experiments on the Main Module: We set the original BiFormer as our baseline. The absence of SSposFFN indicates that the original MLP layers in BiFormer were used. The absence of the spectral decoder and SSDF means we did not use any decoder or cross-layer fusion mechanism in those groups. All experimental results are listed in Table II. Adding any single module enhances the accuracy compared to the baseline, illustrating that each module contributes to the model's performance improvement. In addition, the combination of any two modules gains higher accuracy, with specific exceptions. More importantly, the model with all modules performs the highest accuracy for all datasets, resulting in 8.65%, 4.32%, 3.03%, and 9.03% improvement compared to the baseline model. It demonstrated that the feature extraction on spectral bands in HSIs, combined with the dynamic fusion of spatial and spectral features, significantly enhances the land-cover classification capabilities of the model.

2) Ablation Experiments on the Spectral Decoder: To further verify the effectiveness of our developed SSRA and SiFormer, we designed an additional experiment that replaced the spectral decoder with a decoder of the same structure as the spatial encoder and trained the model as a comparison. The spatial decoder used BRA and spatial position feedforward network, and the SSDF still accepted inputs from the corresponding encoder and decoder. As shown in Table III, the spatial decoder exhibits slightly better accuracy when only the decoder module is utilized. However, the spectral decoder consistently outperformed the spatial decoder across all remaining combinations and datasets, especially when combined with both other two modules. The integrated SSFBF improves the OA of 1.73%, 0.31%, 0.5%, and 0.38% on the IP, SA, HU, and KSC datasets when compared to the substitution. It further validates the capability of SSRA and SiFormer block's contribution to spectral feature extraction.

3) Ablation Experiments on the SSposFFN: We proposed different position feedforward network networks for BRA and SSRA, respectively. To validate their effectiveness, we designed an experiment between SSposFFN and the original position feedforward network (posFFN) from

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TABLE VI
CLASSIFICATION RESULTS OF THE DIFFERENT METHODS ON THE SA DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Class	CNN-based Methods			Transformer-based Methods			CNN-Transformer-based Methods			
	HybridSN	DBDA	CLOLN	MFT	SS1DSwin	3D-SRT	Hit	MF	MASSF	SSFBF
Brocoli green weeds1	99.94 (±0.10)	99.99 (±0.03)	84.71 (±23.93)	99.59 (±0.95)	97.14 (±3.29)	97.80 (±1.41)	99.27 (±0.89)	99.77 (±0.65)	99.85 (±0.10)	**100.00 (±0.00)**
Brocoli green weeds2	99.47 (±0.58)	99.50 (±0.63)	93.02 (±6.24)	99.23 (±0.98)	94.86 (±1.54)	97.83 (±2.71)	97.34 (±1.38)	99.17 (±0.71)	99.57 (±0.55)	**99.91 (±0.24)**
Fallow	**99.08 (±0.86)**	95.45 (±3.04)	90.39 (±7.31)	97.28 (±1.95)	93.04 (±2.10)	87.16 (±4.69)	96.93 (±3.50)	96.47 (±2.88)	99.04 (±1.37)	98.35 (±1.08)
Fallow rough plow	97.71 (±1.40)	96.54 (±1.57)	95.09 (±6.91)	97.07 (±1.92)	98.19 (±0.23)	97.41 (±2.53)	93.67 (±3.31)	97.96 (±1.32)	95.91 (±1.88)	**98.73 (±1.15)**
Fallow smooth	**99.18 (±0.67)**	97.18 (±3.27)	94.49 (±3.57)	97.79 (±2.22)	90.87 (±1.00)	97.19 (±4.97)	96.48 (±1.30)	98.28 (±0.83)	98.53 (±1.03)	99.13 (±0.85)
Stubble	99.91 (±0.12)	**100.00 (±0.01)**	93.78 (±6.22)	99.82 (±0.19)	99.62 (±0.25)	99.38 (±1.13)	99.69 (±0.12)	99.86 (±0.18)	99.32 (±0.37)	99.80 (±0.15)
Celery	**99.97 (±0.07)**	99.92 (±0.12)	59.09 (±34.24)	99.68 (±0.53)	99.58 (±0.17)	99.63 (±0.47)	99.35 (±0.88)	99.44 (±0.41)	99.90 (±0.18)	99.86 (±0.25)
Grapes untrained	96.41 (±0.86)	94.47 (±2.60)	**99.44 (±1.53)**	93.24 (±3.31)	81.58 (±2.32)	83.35 (±2.53)	91.09 (±1.72)	93.77 (±2.29)	97.22 (±0.83)	96.71 (±1.05)
Soil vineyard develop	99.76 (±0.17)	98.51 (±1.13)	67.75 (±20.77)	99.68 (±0.35)	98.93 (±0.63)	99.38 (±0.88)	99.67 (±0.60)	98.85 (±1.44)	99.39 (±0.41)	**99.92 (±0.19)**
Corn senesced green weeds	98.79 (±0.98)	94.37 (±3.76)	93.41 (±4.77)	95.22 (±3.07)	90.46 (±2.89)	94.40 (±4.59)	97.28 (±2.02)	98.15 (±1.52)	97.30 (±1.28)	**99.64 (±0.33)**
Lettuce romaine 4wk	94.01 (±2.67)	**96.25 (±3.13)**	91.30 (±4.34)	94.83 (±3.92)	87.83 (±6.56)	88.49 (±3.86)	90.14 (±3.60)	87.90 (±6.01)	94.76 (±2.80)	95.42 (±1.84)
Lettuce romaine 5wk	99.33 (±0.72)	96.85 (±7.93)	88.93 (±6.48)	98.99 (±1.51)	95.16 (±6.69)	96.90 (±3.05)	98.50 (±0.75)	97.40 (±4.04)	99.11 (±0.88)	**99.34 (±1.29)**
Lettuce romaine 6wk	98.17 (±0.79)	98.16 (±2.52)	95.86 (±3.55)	99.12 (±2.09)	89.90 (±2.10)	95.94 (±1.97)	95.06 (±2.95)	98.60 (±1.14)	99.17 (±0.66)	**99.45 (±0.90)**
Lettuce romaine 7wk	98.52 (±1.14)	99.42 (±0.95)	98.48 (±0.69)	98.51 (±1.54)	86.09 (±7.33)	94.90 (±6.01)	96.29 (±1.15)	**99.44 (±0.36)**	97.45 (±2.21)	98.88 (±0.80)
Vinyard untrained	92.98 (±1.54)	81.75 (±3.94)	91.58 (±6.18)	84.13 (±2.55)	79.13 (±2.36)	78.62 (±3.35)	81.61 (±3.44)	89.69 (±2.12)	94.80 (±1.51)	**94.44 (±1.56)**
Vinyard vertical trellis	**99.83 (±0.37)**	99.63 (±0.41)	89.93 (±3.93)	99.12 (±1.31)	95.92 (±1.04)	97.23 (±1.26)	98.81 (±1.02)	98.70 (±1.09)	99.40 (±0.71)	99.57 (±0.66)
OA	97.79 (±0.31)	94.84 (±0.32)	97.48 (±0.57)	95.40 (±1.21)	90.34 (±1.10)	91.50 (±0.45)	94.15 (±0.57)	96.20 (±0.23)	97.94 (±0.36)	**98.19 (±0.28)**
AA	98.65 (±0.19)	96.52 (±0.87)	97.77 (±0.56)	97.43 (±1.02)	92.53 (±1.13)	94.18 (±0.76)	96.04 (±0.66)	97.12 (±0.47)	98.30 (±0.40)	**98.94 (±0.24)**
k×100	97.54 (±0.35)	94.26 (±0.36)	97.20 (±0.63)	94.88 (±1.35)	89.24 (±1.23)	90.53 (±0.50)	93.49 (±0.63)	95.76 (±0.26)	97.71 (±0.40)	**97.99 (±0.31)**

[43]. In the experiment, both spatial and spectral position feedforward networks were replaced with the original posFFN. The group without a spectral decoder and SSDF only used the spatial position feedforward network for BRA. As presented in Table IV, the SSposFFN outperformed the original position feedforward module except for groups without a spectral decoder and SSDF for the SA dataset. The SSFBF model with all three modules acquired improvements of 1.62%, 0.11%, 0.21%, and 0.29% on IP, SA, HU, and KSC datasets compared to the groups with posFFN and two other modules. Our SSposFFN can better distinguish local spatial and spectral features from BRA and SSRA compared to posFFN and the original MLP layers, highlighting its capacities for aggregating information characterized by high semantic density.

E. Classification Results and Analysis

To analyze the superiority of the proposed SSFBF model, we implemented 9 state-of-the-art algorithms for comparison. We divide them into three groups: CNN-based methods, Transformer-based methods, and CNN–Transformer-based methods. The details can be found in Section III-B.

1) Quantitative Analysis: Tables V and VI–VIII show the quantitative classification results of all the algorithms on the four datasets. All results of the comparison experiments are listed in the form of mean values ± standard deviations, with the optimal results in each row highlighted in bold. The previous methods performed well on specific datasets but yielded poor results on others. Furthermore, SSFBF attained the highest OA and Kappa in IP and the highest OA, AA, and k across all other datasets.

Table V listed the results on the IP dataset. SSFBF outperformed other methods on most land-cover categories, but for some single categories, the other methods were close and even better than SSFBF. Moreover, some CNN-based algorithms, such as HybridSN and CLOLN, earned higher accuracy in OA than most Transformer-based and CNN–Transformer-based algorithms. The predominant reason is that the sample size in the IP dataset is small, and many samples have pool spectral continuity. The local respective fields of CNNs enable them to extract more helpful features compared with the Transformer's long-range dependencies through MHSA. Furthermore, SSFBF can explore more spatial–spectral features through its SSposFFN and fuse them sufficiently through SSDF, leading to its highest OA of 94.17 %, which represents a

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TABLE VII
CLASSIFICATION RESULTS OF THE DIFFERENT METHODS ON THE HOUSTON 2013 DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Class	CNN-based Methods			Transformer-based Methods			CNN-Transformer-based Methods			
	HybridSN	DBDA	CLOLN	MFT	SS1DSwin	3D-SRT	Hit	MF	MASSF	SSFBF
Healthy grass	98.46 (±0.53)	93.57 (±2.65)	95.38 (±1.96)	97.29 (±2.36)	93.95 (±3.66)	89.37 (±4.55)	95.84 (±2.24)	96.78 (±0.91)	92.15 (±4.98)	**97.90 (±0.58)**
Stressed grass	98.63 (±0.66)	97.74 (±1.93)	96.55 (±2.45)	98.34 (±1.50)	95.71 (±2.30)	92.24 (±3.02)	96.62 (±2.10)	99.12 (±0.65)	83.16 (±7.65)	**99.42 (±0.38)**
Synthetic grass	99.98 (±0.05)	99.77 (±0.70)	**100.00 (±0.00)**	99.48 (±0.78)	99.56 (±0.51)	87.74 (±6.78)	99.98 (±0.05)	99.66 (±0.74)	96.13 (±2.57)	**100.00 (±0.00)**
Trees	96.54 (±0.44)	97.24 (±1.08)	96.67 (±1.53)	95.52 (±2.18)	**97.76 (±1.06)**	93.86 (±3.77)	91.81 (±1.54)	97.47 (±1.34)	82.09 (±11.12)	97.51 (±0.52)
Soil	**99.92 (±0.15)**	98.43 (±1.44)	99.43 (±0.57)	99.27 (±0.43)	98.58 (±0.75)	93.44 (±3.36)	99.85 (±0.12)	98.99 (±1.03)	94.90 (±2.75)	99.33 (±0.66)
Water	98.01 (±2.65)	**100.00 (±0.00)**	99.72 (±0.85)	93.61 (±5.50)	96.86 (±2.92)	72.03 (±9.49)	98.32 (±1.53)	97.63 (±2.26)	99.50 (±0.65)	98.70 (±1.17)
Residential	95.22 (±2.49)	91.47 (±2.48)	95.83 (±2.43)	95.61 (±1.59)	86.59 (±2.42)	75.70 (±4.03)	86.93 (±1.69)	**95.98 (±1.56)**	75.38 (±7.20)	95.71 (±1.29)
Commercial	98.17 (±1.28)	99.07 (±0.53)	**99.39 (±1.09)**	97.13 (±1.58)	95.48 (±2.42)	73.05 (±5.45)	95.93 (±1.85)	97.89 (±1.16)	88.70 (±5.37)	98.90 (±0.76)
Road	95.11 (±1.63)	89.27 (±2.95)	**95.56 (±3.73)**	93.15 (±3.20)	84.46 (±1.77)	74.70 (±6.19)	87.11 (±3.85)	94.06 (±1.59)	74.59 (±4.61)	93.93 (±0.9)
Highway	95.10 (±1.51)	85.83 (±3.90)	93.82 (±1.24)	94.60 (±1.49)	85.66 (±1.53)	65.03 (±4.07)	88.97 (±1.58)	**95.32 (±1.67)**	70.97 (±10.21)	94.03 (±1.09)
Railway	98.00 (±1.44)	83.21 (±4.17)	98.00 (±1.37)	95.50 (±2.09)	82.39 (±3.29)	62.10 (±3.60)	92.01 (±2.35)	94.38 (±1.90)	68.16 (±14.76)	**98.62 (±1.07)**
Parking Lot 1	92.63 (±2.40)	87.14 (±4.44)	92.98 (±3.52)	92.59 (±2.36)	88.68 (±1.52)	57.61 (±8.24)	91.34 (±4.71)	94.18 (±2.07)	76.02 (±6.97)	**95.38 (±1.31)**
Parking Lot 2	96.44 (±2.37)	94.50 (±1.96)	94.45 (±1.30)	86.44 (±3.99)	83.26 (±4.03)	74.65 (±5.08)	86.48 (±2.77)	90.98 (±3.81)	80.46 (±6.05)	**95.63 (±2.50)**
Tennis Court	96.70 (±2.47)	97.91 (±2.13)	98.70 (±1.72)	98.62 (±2.75)	98.99 (±0.81)	88.88 (±3.86)	99.53 (±0.58)	99.49 (±0.95)	95.92 (±3.85)	**99.91 (±0.19)**
Running Track	93.64 (±1.00)	**96.91 (±1.35)**	96.08 (±1.76)	94.59 (±2.47)	96.27 (±1.72)	86.92 (±5.65)	91.67 (±0.57)	94.84 (±1.66)	88.18 (±5.28)	93.77 (±0.49)
OA	96.73 (±0.37)	93.02 (±0.61)	96.49 (±0.54)	95.63 (±0.95)	91.49 (±0.70)	77.99 (±1.55)	92.94 (±0.70)	96.37 (±0.41)	81.09 (±2.71)	**97.08 (±0.21)**
AA	96.59 (±0.38)	93.01 (±0.67)	96.50 (±0.46)	95.54 (±1.19)	90.81 (±0.72)	76.77 (±2.09)	92.94 (±0.59)	96.01 (±0.47)	82.65 (±2.35)	**96.70 (±0.26)**
k×100	96.46 (±0.40)	92.46 (±0.66)	96.21 (±0.58)	95.28 (±1.02)	90.80 (±0.76)	76.19 (±1.68)	92.37 (±0.75)	96.07 (±0.44)	79.56 (±2.92)	**96.84 (±0.22)**

significant improvement ranging from 1.68 % to 17.04 % over other methods. Significant improvements were also observed in AA and k.

Tables VI and VII display the results on the SA and HU datasets. The SSFBF also achieved the highest OAs of 98.19% and 97.08%. The best indicators of AA, k, and the accuracy of most land-over categories were also gained by SSFBF, further demonstrating the superiority of SSFBF. It can also be found that the Transformer-based models and CNN–Transformer-based methods have comparable or even better performance in these datasets. The leading cause is that SA and HU contain more samples and exhibit more balanced category distributions. Meanwhile, the spectral continuity of samples in them is well-preserved. Transformer can extract more worthwhile information through long-range dependencies. Furthermore, the proposed SSFBF model fully uses the local strategy of SSposFFN and the global strategy of BRA and SSRA in the spatial and spectral domains, dramatically improving performance.

Table VIII presents the results on the KSC dataset. SSFBF achieved a definite advantage on all three evaluation metrics and most land-cover categories, obtaining the highest OA of 98.25%. CLOLN earned the second-highest OA of 95.31%, whereas HybridSN exhibited the worst OA of 71.77%. The results in KSC were significantly different from those in previous datasets. Neither CNN nor Transformer-based methods demonstrated strong performance. The explanation is that the samples of the KSC dataset are scattered, with few overlapping relationships among them. However, the SSFBF can still perform best, demonstrating its superiority in classifying datasets under varying conditions.

2) Qualitative Analysis: To further intuitively present the performance, we illustrated classification maps of SSFBF and other methods on the IP dataset. As shown in Fig. 5, the plot quality of SSFBF is the best, corresponding to its quantitative results. The CNN-based models produce more misclassification of boundary pixels between land-cover categories, likely explained by CNN's limited reception fields that can only partially consider edge information. The Transformer-based methods misclassify more inner pixels into specific categories, attributed to the Transformer's capacity to model global dependencies, often leading to neglecting some local details. Therefore, better results could be observed in models that combine CNNs and Transformers, such as MF and MASSF.

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TABLE VIII
CLASSIFICATION RESULTS OF THE DIFFERENT METHODS ON THE KENNEDY SPACE CENTER DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

	CNN-based Methods			Transformer-based Methods			CNN-Transformer-based Methods			
Class	HybridSN	DBDA	CLOLN	MFT	SS1DSwin	3D-SRT	Hit	MF	MASSF	SSFBF
Scrub	62.38 (±4.20)	81.93 (±8.25)	**99.45 (±0.76)**	85.74 (±7.50)	99.13 (±0.35)	97.26 (±3.65)	97.50 (±1.65)	92.16 (±5.57)	93.79 (±5.48)	99.39 (±0.75)
Willow swamp	66.58 (±9.37)	82.19 (±11.15)	96.58 (±2.38)	66.16 (±10.35)	88.34 (±6.33)	81.17 (±13.04)	96.44 (±1.59)	79.69 (±6.81)	65.85 (±8.69)	**98.67 (±1.42)**
CP hammock	48.42 (±5.67)	78.46 (±11.89)	92.43 (±12.14)	69.98 (±16.80)	80.77 (±11.73)	74.24 (±16.10)	82.37 (±6.02)	88.46 (±15.92)	57.69 (±7.38)	**92.35 (±4.73)**
Slash pine	64.60 (±6.68)	78.09 (±9.09)	75.06 (±10.32)	89.22 (±8.54)	73.68 (±3.53)	59.07 (±9.60)	80.03 (±7.00)	75.94 (±9.65)	54.10 (±6.59)	**93.96 (±2.23)**
Oak/Broadleaf	64.60 (±4.36)	87.19 (±12.41)	84.43 (±10.13)	88.35 (±12.91)	61.77 (±9.03)	62.14 (±11.81)	85.18 (±7.71)	78.15 (±9.76)	77.00 (±9.30)	**96.52 (±3.13)**
Hardwood	69.61 (±28.01)	76.16 (±20.49)	82.05 (±11.68)	57.30 (±14.11)	78.09 (±7.98)	74.30 (±13.04)	90.01 (±2.85)	75.58 (±10.33)	76.48 (±11.58)	**91.03 (±1.96)**
Swap	10.00 (±30.00)	86.85 (±29.47)	94.39 (±10.52)	79.29 (±39.67)	89.74 (±8.07)	80.44 (±16.17)	**99.41 (±0.79)**	98.56 (±2.56)	82.14 (±14.59)	97.15 (±2.58)
Graminoid marsh	61.28 (±3.30)	71.49 (±5.80)	91.14 (±6.10)	72.12 (±11.79)	90.72 (±3.35)	89.41 (±4.99)	96.46 (±1.44)	77.01 (±10.68)	74.97 (±6.52)	**97.45 (±1.42)**
Spartina marsh	38.83 (±4.29)	97.19 (±3.49)	97.71 (±3.66)	90.73 (±10.64)	97.49 (±1.59)	94.42 (±4.33)	**99.76 (±0.19)**	98.12 (±2.72)	90.27 (±6.11)	99.43 (±0.79)
Cattail marsh	78.61 (±5.11)	**100.00 (±0.00)**	98.95 (±2.86)	92.94 (±5.42)	98.84 (±1.11)	92.12 (±7.88)	90.89 (±4.38)	93.36 (±4.10)	66.22 (±7.40)	99.57 (±0.57)
Salt marsh	94.13 (±2.60)	**100.00 (±0.00)**	**100.00 (±0.00)**	97.25 (±2.92)	98.14 (±1.70)	98.62 (±1.50)	99.76 (±0.52)	97.28 (±3.87)	91.21 (±2.51)	**100.00 (±0.00)**
Mud flats	83.19 (±2.04)	98.05 (±1.47)	98.93 (±1.14)	90.14 (±3.53)	94.50 (±1.93)	91.99 (±4.93)	87.65 (±2.20)	88.00 (±4.09)	58.25 (±5.12)	**99.93 (±0.16)**
Water	94.23 (±1.79)	**100.00 (±0.00)**	99.95 (±0.12)	98.80 (±1.10)	**100.00 (±0.00)**	97.42 (±1.97)	96.70 (±1.98)	99.32 (±0.47)	78.17 (±2.73)	99.80 (±0.39)
OA	71.11 (±0.96)	89.03 (±1.35)	95.31 (±0.87)	85.29 (±4.04)	92.86 (±1.32)	88.90 (±1.30)	93.86 (±1.28)	89.72 (±2.91)	76.29 (±2.45)	**98.25 (±0.21)**
AA	57.91 (±1.34)	79.45 (±2.78)	92.05 (±1.30)	73.53 (±6.47)	88.07 (±2.06)	82.62 (±2.41)	91.10 (±1.85)	82.73 (±4.79)	67.28 (±3.43)	**97.17 (±0.37)**
k×100	67.49 (±1.00)	87.72 (±1.52)	94.78 (±0.96)	83.54 (±4.52)	92.05 (±1.46)	87.63 (±1.45)	93.15 (±1.43)	88.52 (±3.26)	73.40 (±2.77)	**98.06 (±0.23)**

It is noticeable that the proposed SSFBF can produce more refined visual effects with less noise and misclassification, yielding more reasonable maps than other algorithms.

3) Visualization: To further visually compare the performance of SSFBF and comparison models, we adopted the t-distributed stochastic neighbor embedding (t-SNE) to project the features onto 2-D coordinates intuitively. Fig. 6 displays the 2-D plots on the IP dataset, where dots of different colors represent samples of different categories. The clustering of samples within the same color indicates the effectiveness in classifying the category. We excluded the background category to avoid unnecessary interference. The SSFBF demonstrated superior clustering results, with only a few misclassifications in the plots. In contrast, sample distributions from other methods were relatively dispersed. For models with comparatively poorer performance, such as DBDA and 3D-SRT, samples of classes such as "Grass-pasture-mowed" and "Oats" did not exhibit interclass variance. It further demonstrates our SSFBF's excellent classification performance.

Furthermore, we evaluated the confusion matrices of SSFBF across all four datasets. As shown in Fig. 7, only a few samples in IP were misclassified, primarily due to the imbalance in sample distribution. The classification results for most categories are closely aligned with the ground truth along the diagonal, indicating that most samples are accurately predicted. This demonstrates the superiority of SSFBF.

TABLE IX
AVERAGE TRAINING TIMES AND TESTING TIMES FOR ALL METHODS ACROSS ALL FOUR DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Methods	Training time(s)				Testing time(min)			
	IP	SA	HU	KSC	IP	SA	HU	KSC
HybridSN	1.23	1.37	1.41	1.14	0.23	1.26	5.88	3.02
DBDA	2.60	0.52	2.15	2.35	1.04	5.55	5.81	6.04
CLOLN	0.12	**0.13**	**0.13**	**0.10**	0.06	**0.32**	1.08	0.61
MFT	**0.11**	0.17	**0.13**	0.12	**0.04**	1.13	0.97	0.58
SS1DSwin	0.73	1.16	1.41	0.95	0.09	0.75	**0.18**	**0.06**
3D-SRT	1.15	1.34	1.72	1.37	0.27	1.38	8.53	4.03
Hit	0.72	0.08	0.81	0.62	0.20	0.72	4.35	2.05
MF	0.22	0.25	0.29	0.23	0.08	0.43	2.14	1.16
MASSF	0.12	0.15	0.14	0.12	0.06	0.35	1.19	0.69
SSFBF	1.04	1.14	1.53	1.13	0.21	1.09	6.53	3.08

F. Efficiency Analysis

To explore the effectiveness of the proposed SSFBF, we present the training time and testing time of all methods on four datasets. We used the average training time per epoch and average testing time from ten repeated experiments. As shown in Table IX, the SSFTT performed best on all four datasets. Meanwhile, the CLOLN, MFT, and SS1DSwin exhibited shorter training and testing times than the remaining models, which can be attributed to their relatively simple architecture, lacking hierarchical structures. Nevertheless, the simplicity also results in suboptimal classification accuracy.

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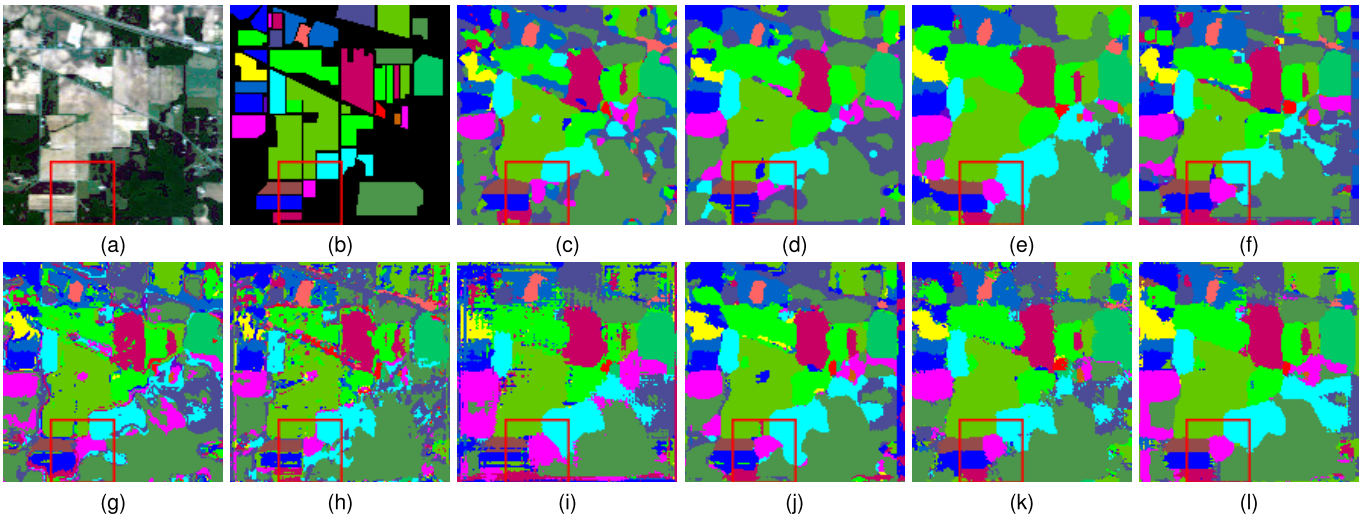


Fig. 5. Classification maps of the India Pines dataset. (a) False-color map, (b) test ground truth, (c) HybridSN (OA = 89.60%), (d) DBDA (OA = 87.69%), (e) CLOLN (OA = 92.49%), (f) MFT (OA = 91.23%), (g) SS1DSwin (OA = 82.12%), (h) 3D-SRT (OA = 77.13%), (i) Hit (OA = 81.82%), (j) MF (OA = 91.74%), (k) MASSF (OA = 92.42%), and (l) SSFBF (OA = 94.17%).

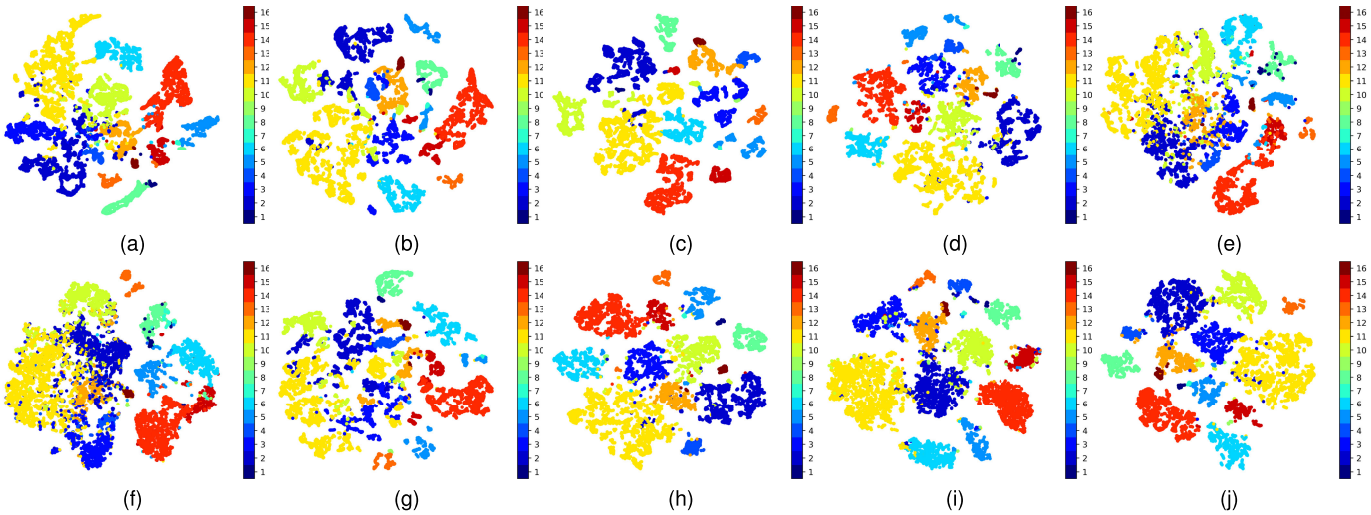


Fig. 6. T-SNE maps of different methods on the India Pines dataset. (a) HybridSN (OA = 89.60%), (b) DBDA (OA = 87.69%), (c) CLOLN (OA = 92.49%), (d) MFT (OA = 91.23%), (e) SS1DSwin (OA = 82.12%), (f) 3D-SRT (OA = 77.13%), (g) Hit (OA = 81.82%), (h) MF (OA = 91.74%), (i) MASSF (OA = 92.42%), and (j) SSFBF (OA = 94.17%).

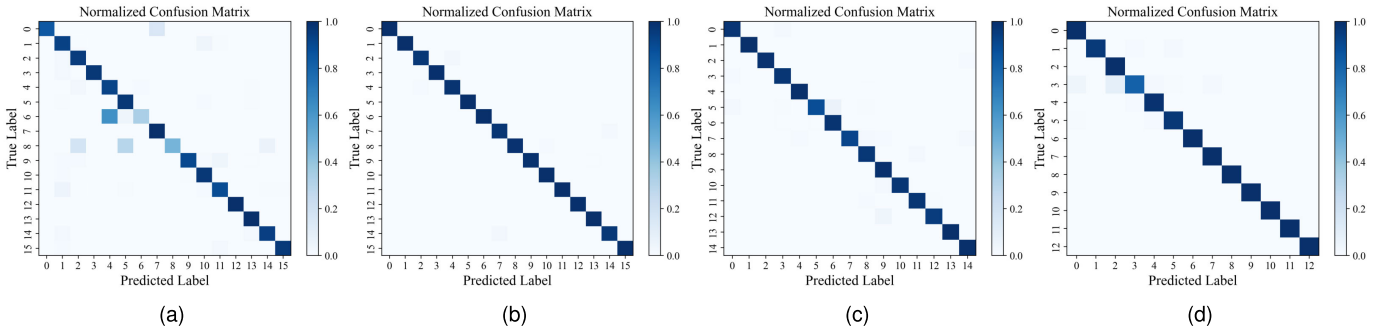


Fig. 7. Confusion matrices of the SSFBF across all four datasets. (a) IP. (b) SA. (c) Houston 2013. (d) KSC.

In contrast, our SSFBF model exhibits more extended training and testing time, which can be explained by SSFBF adopting large patches alongside deeper hierarchical structures and retaining many original channels. However, it achieves state-of-the-art accuracy among all methods.

G. Influence of Training Samples

To validate the generalization capability of SSFBF, we investigated the impact of various training samples on the models' classification results. We selected 20%, 40%, 60%, 80%, and 100% of the original training samples as

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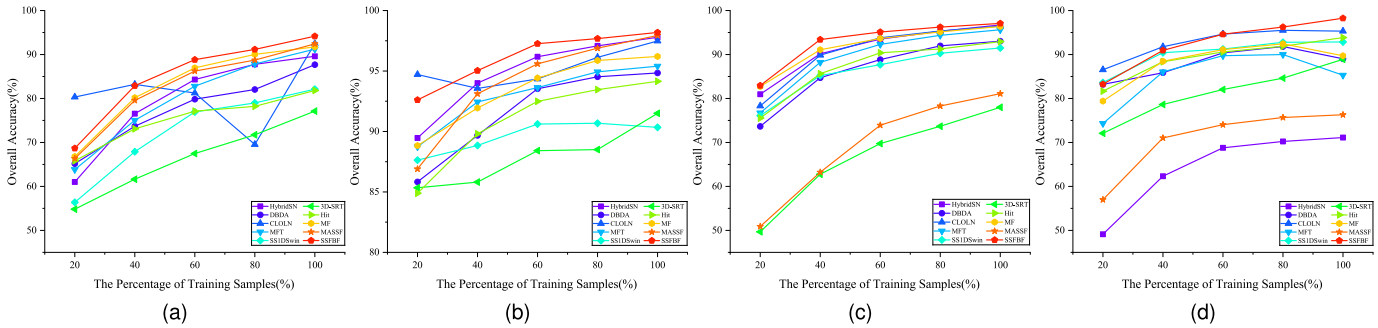


Fig. 8. Overall accuracy of methods across different training sample proportions for all four datasets: (a) IP, (b) SA, (c) Houston 2013, and (d) KSC.

new training samples on four datasets across all methods. The validation set stayed unchanged, and the remaining samples were allocated to the test set.

As shown in Fig. 8, SSFBF demonstrates significant superiority across all datasets. Specifically, SSFBF achieves the highest OA among all methods on the HU and consistently outperforms other methods on the SA except when the percentage is 20%. On the IP and KSC, SSFBF's OA is only lower than CLOLN's at percentages of 20%, 40%, and 60%. At the same time, its performance advantage increases with the size of the training samples, showing a significant advantage when the sample size reaches 100%. The results indicate that the proposed method is more effective and robust.

IV. Conclusion

In this paper, we proposed a novel CNN–Transformer-based method for HSI classification called SSFBF. The BRA and SSRA with dynamic routing attention extract high-resolution spatial relationships and discriminative spectral semantic information through routing mechanisms. An SSposFFN is followed to mitigate local feature loss induced by attention. A novel dynamic fusion mechanism is designed to integrate the acquired spatial–spectral feature, gaining further output of feature representations. Experimental results on four mainstream HSI datasets have demonstrated that our SSFBF achieves a comprehensive feature extraction from all kinds of HSI datasets and remarkable performance compared with several state-of-the-art HSI classification methods.

In the future, we will explore lightweight architectures and experiment with semi-supervised and unsupervised learning techniques to address the scarcity of labeled data in HSI processing.

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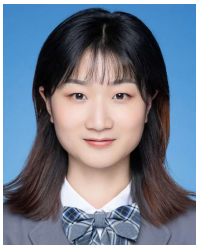
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